

Scenario 1: Image Classification Model Overfits Quickly:

You're training a CNN to classify dog breeds. You achieve 98% accuracy on training data within 5 epochs, but validation accuracy stays at 60%, and test performance is poor.

Answer:

- Use **Data Augmentation** to create more varied training images.
 - Add **Dropout** to reduce over-dependence on specific features.
 - Use **Early Stopping** to stop training when validation performance stops improving.
 - Add **L2 Regularization** to control model complexity.
 - Use a **smaller CNN model** if the model is too complex.
 - Increase the **training dataset** if possible.
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Scenario 2: ANN Fails to Learn Non-Linear Patterns

Project: Simple digit classification using **Artificial Neural Network (ANN)**

Problem: Accuracy stuck around 50%; adding more epochs doesn't improve performance.

Answer:

- Use preprocessing techniques like **normalization**.
 - Change **hyperparameters** such as learning rate, batch size, and number of neurons.
 - Use a **non-linear activation function** like ReLU in hidden layers.
 - If the problem is complex, use a **deeper ANN**.
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Scenario 3: CNN Model Predicts Same Class for All Images

Project: Face Mask Detection using **CNN**

Problem: Model always predicts "With Mask" for every test image.

Answer:

- Use **preprocessing techniques like normalization**.
 - Use **Dropout** to reduce overfitting.
 - Increase the **training dataset** with diverse images of both **with mask and without mask**.
 - Check for **class imbalance** in the dataset.
 - Tune **hyperparameters** such as learning rate and batch size.
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Scenario 4: Object Detection Model Detects Objects but Bounding Boxes Are Misaligned

Project: YOLOv8 used for Tiger Detection in Forest

Problem: Bounding boxes do not fit objects properly.

Answer:

- Check whether the **bounding-box annotations are correct** in the training dataset.
 - Make sure the images and labels are **properly formatted and aligned**.
 - Increase the **training dataset** with different tiger images, angles, and sizes.
 - Use **data augmentation** such as scaling, cropping, and flipping.
 - Tune **YOLOv8 hyperparameters** and train for more epochs if needed.
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Scenario 5: Face Mask Detection Gives High Accuracy but Performs Poorly on Real Faces

Project: Real-time Face Mask Detection using CNN

Problem: High accuracy on test set, but poor real-time performance on webcam feed.

Answer:

- Train with **more real-world images** similar to webcam conditions.
 - Include different **lighting, backgrounds, face angles, and camera qualities**.
 - Use **data augmentation** to create more variations.
 - Check whether the **test dataset is too similar to the training data**.
 - Tune **hyperparameters** such as learning rate and batch size.
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Scenario 6: CNN-Based Object Detector Struggles at Night

Project: YOLOv8-based object detection in CCTV

Problem: Model doesn't detect objects in dark/night images.

Answer:

- Train the model with more **night-time/dark images**.
 - Use **data augmentation** such as brightness and contrast changes.
 - Apply **image preprocessing** to improve low-light images.
 - Make sure the training dataset contains **both day and night images**.
 - Tune **hyperparameters** if needed.
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